Housing Allocation, Fertility, and Population

An illustrative model

Working note for Tommaso De Santo September 6, 2026

Borrowing limits can keep housing with old owners even when young households would give up more consumption to use it. We show that a small reallocation can then improve welfare while holding fertility fixed. We also study housing policy during a demographic transition: an earlier decline in the desire for children starts the adjustment, and a later policy changes its course.

1 Environment

Households. At date t, the economy contains Y_t young households and O_t old households. A young household has income $y_i > 0$, liquid wealth $b_i > 0$, and chooses total nondurable expenditure c, total housing h, completed fertility $n > 0$, next-period net financial wealth a', and tenure $m \in \{R, O\}$. Fertility is a continuous choice made once. The pair (y_i, b_i) is drawn from the same distribution F in each cohort.

Preferences. Each child uses $\chi > 0$ units of goods and $\kappa > 0$ units of housing. Define the parts of the household bundle available to the adults as

$$x = c - \chi n > 0, \qquad s = h - \kappa n > 0. \tag{1}$$

Young and old flow utilities are

$$u_t^y(c,h,n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n, \tag{2}$$

$$u^o(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e. \tag{3}$$

Both child requirements reduce the bundle left for adults. All log arguments and preference weights are positive. Households discount future utility by $\beta > 0$. The estate e consists of financial assets and the proceeds from selling retained housing at death, at the end of old age. The weight ω_B measures how much households value the estate they leave. Estates do not enter their children's initial wealth in this model.

Each household draws once an idiosyncratic preference ξ_i for ownership, independently of (y_i, b_i), and observes it before making its choices. The draw is logistic with location $\bar{\xi}$ and scale $\sigma_\xi > 0$.

Demography. A child born at t produces $\nu > 0$ young households at $t+1$, after survival and household formation. Thus

$$Y_{t+1} = \nu \bar{n}_t Y_t, \qquad O_{t+1} = Y_t, \tag{4}$$

where $\bar{n}_t$ is average fertility among the young. Population counts adult households; Appendix A.3 gives the conversion to resident persons.

Housing and asset markets. The fixed housing stock is $\bar{H}$. Rental units satisfy $h \le h_R^{\max}$, owner units satisfy $h \le h_O^{\max}$, and $h_R^{\max} < h_O^{\max}$. The consumption good is the numeraire. Goods and one-period bonds trade with the rest of the world. The bond price is $q \in (0,1)$ and its gross return is $R_f = q^{-1}$; housing clears against the domestic stock. The owner of record pays property tax at rate τ^p. A competitive rental intermediary buys a unit for $P_t > 0$ at the beginning of date t, then collects r_t, pays $\tau^p P_t$, and holds a unit worth P_{t+1} at the end of the period. Free entry therefore imposes

$$r_t + P_{t+1} = R_f P_t + \tau^p P_t \quad \Longleftrightarrow \quad r_t = (R_f + \tau^p) P_t - P_{t+1}. \tag{5}$$

Rent r_t per unit of housing is paid at the end of period t; its cost at the beginning of the period is $qr_t > 0$. At constant house prices, $r = (R_f - 1 + \tau^p)P$. The government distributes property-tax revenue in equal rebates to young and old households. Let T_t denote each household's rebate, valued at the beginning of the period.

Home finance. A young buyer finances the share $\phi_t \in (0,1)$ and posts $(1-\phi_t)P_t h$ at purchase. Closing occurs before current income and rebates are available, and unsecured borrowing is unavailable. Thus only liquid wealth b_i can satisfy the down payment. The owner cannot buy more housing when old or rent its retained housing to another household. At death the estate sells it back into the fixed stock, and the proceeds enter warm-glow estate expenditure.

2 Household problems

Households take prices and transfers as given. In both tenures, a' denotes financial wealth on entering old age, net of any mortgage repayment.

Young renters. Conditional on renting, the young household solves

$$\begin{aligned} W_t^R(i) = \max_{c,h,n,a'} \; & u_t^y(c,h,n) + \beta V_{t+1}^R(a') \\ \text{s.t.} \quad & c + qa' + q\, r_t h = y_i + b_i + T_t, \\ & h \le h_R^{\max}, \qquad a' \ge 0. \end{aligned} \tag{6}$$

The renter sets aside qa' to enter old age with financial wealth a'.

Young owners. Conditional on owning, the young household solves

$$\begin{aligned} W_t^O(i) = \max_{c,h,n,a'} \; & u_t^y(c,h,n) + \beta V_{t+1}^O(a',h) \\ \text{s.t.} \quad & c + qa' + (1 + q\tau^p)P_t h = y_i + b_i + T_t, \\ & (1-\phi_t)P_t h \le b_i, \qquad h \le h_O^{\max}, \\ & qa' + \phi_t P_t h \ge 0. \end{aligned} \tag{7}$$

The owner sets aside $qa' + \phi_t P_t h$ in bonds. After subtracting mortgage debt, its old-age financial wealth is a', alongside title $H = h$. The last constraint requires nonnegative bond holdings; a' itself can be negative. Below, positive saving means positive bond holdings, before subtracting mortgage debt. The property tax is also set aside in a bond.

Old households. An old household has financial wealth a. An owner also carries housing H from young age and retains $h^2 \le H$ for use while old. Let $a^e \ge 0$ denote financial saving during old age. For a household old at date t, the estate left at death is

$$e = \begin{cases} R_f a^e, & \text{renter}, \\ R_f a^e + P_{t+1} h^2, & \text{owner}. \end{cases} \tag{8}$$

Financial saving earns the bond return, and the owner's retained housing is sold at the end of old age. Housing sold at the beginning of old age, $H - h^2$, instead provides current funds. The owner's budget before substituting for financial saving is

$$c^2 + a^e + q\tau^p P_t h^2 = a + P_t(H - h^2) + T_t. \tag{9}$$

Substituting $a^e = q(e - P_{t+1}h^2)$ and using (5) gives the owner's budget below.

Renters and owners solve, respectively,

$$\begin{aligned} V_t^R(a) &= \max_{c^2,h^2,e} u^o(c^2,h^2,e), \\ &\quad c^2 + qe + q\,r_t h^2 = a + T_t, \quad 0 < h^2 \le h_R^{\max}, \end{aligned} \tag{10}$$

$$\begin{aligned} V_t^O(a,H) &= \max_{c^2,h^2,e} u^o(c^2,h^2,e), \\ &\quad c^2 + qe + q\,r_t h^2 = a + P_t H + T_t, \quad 0 < h^2 \le H, \quad e \ge P_{t+1}h^2. \end{aligned} \tag{11}$$

The owner's estate bound $e \ge P_{t+1}h^2$ is equivalent to nonnegative financial saving $a^e \ge 0$.

Tenure choice. A household owns when $W_t^O(i) + \xi_i \ge W_t^R(i)$. The logistic ownership taste gives the owner share

$$\pi_t^O(i) = \frac{\exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}{\exp\{W_t^R(i)/\sigma_\xi\} + \exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}. \tag{12}$$

The taste draw smooths aggregate tenure shares without changing conditional renter or owner policies.

3 Equilibrium

Prices and transfers equate housing demand to the fixed stock and balance the government budget. Let $\bar{h}_t^Y$ and $\bar{h}_t^O$ denote average housing per young and old household, averaging over types and tenure choices. Let G_t be the old cohort's normalized distribution over net financial wealth, inherited housing titles, and tenure, with separate mass O_t. Housing clearing and the government budget are

$$Y_t\bar{h}_t^Y + O_t\bar{h}_t^O = \bar{H}, \qquad (Y_t + O_t)T_t = q\tau^p P_t \bar{H}. \tag{13}$$

The second condition rebates all property-tax receipts in current value. Goods and bonds trade externally; their markets do not impose an additional domestic clearing equation.

Definition 1 (Equilibrium)**.** *Given policies, fertility preferences, the initial state (Y_0, O_0, G_0), and the fixed entrant distribution F, an equilibrium consists of price and transfer sequences (P_t, r_t, T_t), household choices, tenure probabilities, and cohort states (Y_t, O_t, G_t) satisfying:*

(i) Young and old households solve their stated problems, and tenure follows (12).

(ii) Rental entry satisfies (5).

(iii) Housing clears and the government budget balances as in (13).

(iv) Cohort sizes follow (4); G_{t+1} is the distribution of assets, titles, and tenure chosen by the young at t.

After an unexpected reform, old households keep their inherited assets and titles and reoptimize using the new price path. Existing contractual obligations remain fixed.

4 Housing allocation

We ask whether the planner can improve the use of the existing housing stock by moving space from old owners to young households. The planner chooses housing and consumption directly. It respects the physical limits on each tenure and can relax private financing restrictions. Fertility, tenure, estates, and all future real allocations remain fixed.

Let i be a young owner and j an old owner. Their marginal values of housing, measured in units of current consumption, are:

$$MV_i^Y = \frac{\alpha(c_i - \chi n_i)}{h_i - \kappa n_i} = \frac{\alpha x_i}{s_i}, \qquad MV_j^O = \frac{\gamma c_j^2}{h_j^2}. \tag{14}$$

These values measure the consumption each household would give up for a little more housing. Comparing them requires no interpersonal comparison of utility levels.

Proposition 1 (Housing misallocation)**.** *Suppose matched groups of young and old owners have equal positive mass and room for a small transfer of housing. Assume uniformly positive young saving, adult consumption, and room below the young physical housing cap. If $MV_i^Y - MV_j^O$ is uniformly positive on matched pairs, a sufficiently small reallocation toward the young can make them better off and leave every other household equally well off, in the absence of real reallocation costs.*

Proof. Give the young owner ϵ units of the old owner's housing. Keeping the old owner's utility fixed requires additional consumption:

$$D_j(\epsilon) = c_j^2 \left[\left(\frac{h_j^2}{h_j^2 - \epsilon} \right)^{\gamma} - 1 \right]. \tag{15}$$

Take this amount from the young owner. Goods and housing balance. Since $D_j'(0) = MV_j^O$, the young owner's utility gain has derivative $(MV_i^Y - MV_j^O)/x_i > 0$ at zero. A small change preserves feasibility. The seller replaces the lost estate title with a bond, and the buyer's later sale of the extra title repays its financing. These claims offset, so future allocations and existing creditors are unchanged. Appendix A.1 gives the payments. □

Why the values can differ. With positive saving and interior old-age choices, a young owner who would buy more housing if its down-payment limit were relaxed has $MV_i^Y > qr_t$, provided the physical owner cap is slack. An old owner with slack retention and estate bounds instead satisfies:

$$MV_j^O = \frac{\gamma c_j^2}{h_j^2} = qr_t. \tag{16}$$

Young buyers need cash at purchase; old owners already hold their homes. The resulting value gap gives the reallocation in Proposition 1. The argument applies at any date when these conditions hold, including during a demographic transition.

The rental size limit restricts an alternative way to obtain space. A household can be unable to rent a larger home and unable to fund its purchase. The owner-to-owner comparison above holds tenure fixed; a renter already at its physical cap cannot receive more housing in that tenure. An ownership taste can make constrained ownership preferable even when larger rentals are available.

If moving ϵ housing uses $C(\epsilon)$ goods, with $C(0) = 0$ and $C'(0) = K_{ijt} \geq 0$, the condition becomes $MV_i^Y > MV_j^O + K_{ijt}$. The young pays this real cost as well as the old owner's compensation. A fixed moving cost requires a finite gain large enough to cover it.

The proposition establishes inefficiency relative to a planner who can relax private finance. It identifies a direction of improvement without solving the entire first-best allocation. A market-based reform with households choosing freely requires a separate welfare comparison.

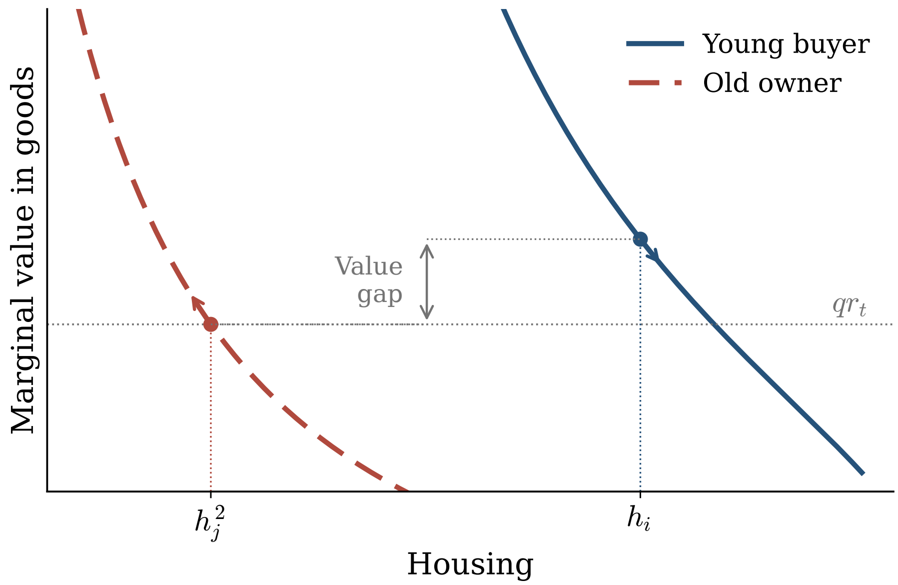

Figure 1: Housing moves from the old owner to the young buyer, with the old owner compensated in consumption. Each curve uses the household's own housing on the horizontal axis. The arrows describe the same transfer of space. Fertility and future real allocations are fixed.

5 Housing access and fertility

More space makes children easier to accommodate. Paying for that space reduces the goods available to the family. Fertility rises when the first effect dominates the second.

Consider the young owner's conditional problem at a given date. Assume positive saving, a binding down payment, a slack physical housing cap, and interior old-age choices. Suppress household and date indices. Let $k = \beta(1 + \gamma + \omega_B)$ and $w = y + b + T_t + qT_{t+1}$, lifetime resources before housing. Eliminating saving from the original problem gives:

$$(1 + k)x + \chi n + uh = w, \qquad u = qr_t. \tag{17}$$

The net cost of housing is u per unit, after accounting for its eventual sale. The fertility first-order condition is:

$$\frac{\vartheta}{n} = \frac{\chi}{x} + \frac{\alpha\kappa}{s}, \qquad s = h - \kappa n. \tag{18}$$

The marginal benefit of a child equals its goods and space requirements, valued by the household.

Let p be the net lifetime payment for a marginal unit of extra current housing, including any accompanying transfers. With prices and other resources fixed, $p = u$; compensation or a subsidy changes that payment. The budget response is $(1 + k)\mathrm{d}x = -\chi\mathrm{d}n - p\mathrm{d}h$. The implied fertility response is:

Proposition 2 (A fertility condition)**.** *Under these household conditions, more housing increases fertility locally if and only if:*

$$\frac{\alpha\kappa}{(h - \kappa n)^2} > \frac{\chi p}{[1 + \beta(1 + \gamma + \omega_B)](c - \chi n)^2}. \tag{19}$$

The comparison holds at any date, while tenure and the binding constraints remain unchanged and saving and old-age choices are reoptimized.

The condition uses household allocations and the net payment, with no borrowing multiplier. It is a conditional prediction of the household problem; a policy's equilibrium effect also includes changes in prices, transfers, and tenure. Appendix A.2 gives a sufficient restriction using only parameters in a stationary specialization, and its extension to dates with nonincreasing expected house prices. The fixed-fertility welfare comparison in Proposition 1 does not require this sign.

6 Fertility decline and policy during the transition

The economy need not be in a steady state when policy changes. Consider an initial steady state with fertility weight ϑ_0 and financed share ϕ_0. At date 0 an unexpected permanent fall to $\vartheta_1 = \vartheta_0 - \epsilon$, with $\epsilon > 0$, starts a baseline transition. Explaining this initial change in preferences is outside the exercise.

At a later date $t_p \geq 1$, compare continued policy ϕ_0 with a permanent increase to $\phi_1 = \phi_0 + \delta$, where $\delta > 0$. The increase is announced and implemented at t_p. The two continuations therefore start from the same young and old cohorts and the same inherited assets and housing titles. They face the same ϑ_1 and all other non-policy primitives. This construction represents a housing policy introduced during an adjustment already in progress.

The unanticipated policy may cause prices to jump at t_p, while inherited assets and titles remain fixed. With earlier announcement, the two paths would begin to differ at that date.

6.1 Steady states and population

For constant prices and policies, the household problems determine mean fertility and housing at any candidate (P, T). The old distribution comes from repeating the young household choices. A positive steady state solves:

$$\bar{n}(P, T; \vartheta, \phi) = \frac{1}{\nu}, \qquad T = \frac{q\tau^p P}{2}\,[\bar{h}^Y(P, T) + \bar{h}^O(P, T)]. \tag{20}$$

The first condition is replacement; the second rebates property-tax revenue. Once a positive root exists, housing clearing determines population:

$$N^*_{\text{hh}} = Y^* + O^* = 2Y^* = \frac{2\bar{H}}{\bar{h}^{Y*} + \bar{h}^{O*}}. \tag{21}$$

Thus two steady states can have the same replacement fertility and different population levels. With the fixed stock, a larger population uses less housing per household on average.

In the mixed-tenure economies used below, a small fall in ϑ lowers stationary price and population. Housing per young and per old household rises. A small increase in ϕ raises stationary price, population, and young housing; old housing falls by more than young housing rises. These are local comparisons around an economy with both renters and owners.

6.2 The two continuations

Let superscripts 0 and 1 denote the baseline and policy paths from t_p. Because their initial young cohorts are identical, the demographic law gives:

$$\frac{Y^1_{t_p+k}}{Y^0_{t_p+k}} = \prod_{j=0}^{k-1} \frac{\bar{n}^1_{t_p+j}}{\bar{n}^0_{t_p+j}}, \qquad O^a_{t+1} = Y^a_t \quad (a = 0, 1). \tag{22}$$

Relative cohort size records the accumulated fertility differences. This identity is useful along the entire transition, without assuming that either path is stationary.

Proposition 3 (A decline followed by a housing policy)**.** *There is an open set of economies with positive child goods costs, positive rebated property tax, and positive masses of renters and owners for which the following holds. A sufficiently small permanent fall in ϑ produces a converging baseline transition with lower initial fertility and a smaller terminal population. A sufficiently small permanent increase in ϕ, introduced at any later date on that baseline, has a converging continuation from the same inherited state and satisfies:*

$$\bar{n}^1_{t_p} > \bar{n}^0_{t_p}, \qquad N^*_{\mathrm{hh},1} > N^*_{\mathrm{hh},0}. \tag{23}$$

The same constraints continue to bind along these local transitions.

Appendix B gives the assumptions and proof. The policy need not restore the original population or raise fertility above replacement.

Both terminal fertility levels equal $1/\nu$. With exponential convergence, (22) also implies:

$$\sum_{t=t_p}^{\infty} \log \frac{\bar{n}^1_t}{\bar{n}^0_t} = \log \frac{N^*_{\mathrm{hh},1}}{N^*_{\mathrm{hh},0}} > 0. \tag{24}$$

The cumulative relative fertility effect is positive. Individual dates can have effects of either sign: the example admits damped oscillations as it approaches the new steady state.

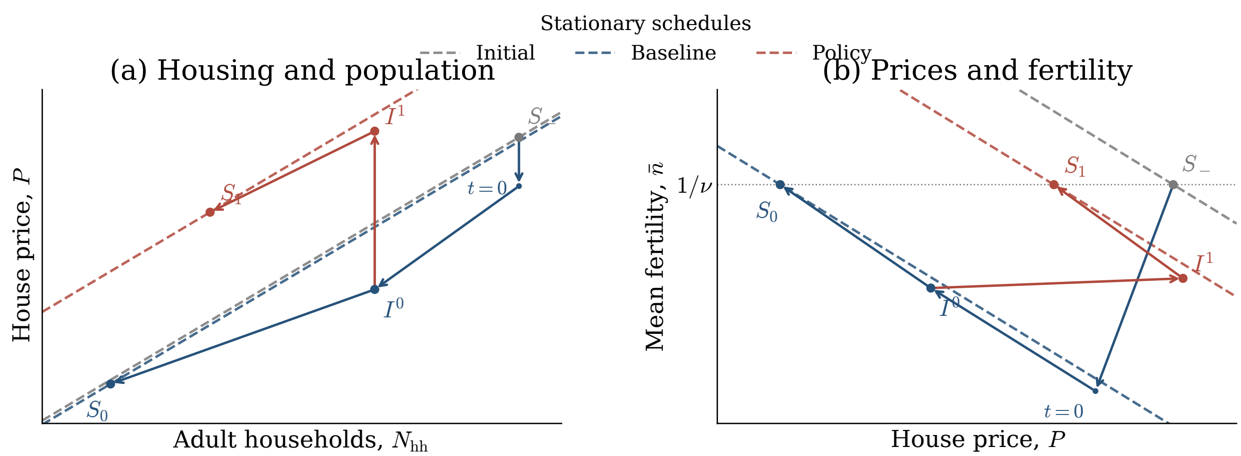

Figure 2: A fertility decline followed by a housing policy. From the old steady state S_{-}, the preference decline at $t = 0$ starts the move toward S_0. The policy changes the intervention-date equilibrium from I^0 to I^1, with the same inherited cohorts and claims, and leads to S_1. Dashed lines are constant-price schedules. Arrows connect selected first-order equilibrium responses and do not imply monotone adjustment. Appendix B.4 defines the comparison.

The permanent policy changes the terminal household problem. A temporary reform that restores all final primitives cannot change a unique stationary endpoint if both paths converge to it. The population comparison also does not establish a welfare gain from the credit reform. The compensated allocation and the competitive policy answer different questions.

A Household arguments

A.1 Financial settlement of the reallocation

The direct-allocation proof uses consumption and housing as resources. Prices enter only to check that inherited claims and future allocations can be respected. Set $u_t = qr_t$ and write the payment to the seller in excess of the title price as:

$$L(\epsilon) = D_j(\epsilon) - u_t\epsilon. \tag{25}$$

The seller receives $P_t\epsilon + L(\epsilon)$ and releases the tax reserve $q\tau^p P_t\epsilon$. After consuming $D_j(\epsilon)$, it buys a bond costing $qP_{t+1}\epsilon$, replacing the title lost from its estate. The young buyer spends $D_j(\epsilon)$ less on current consumption and finances the rest of the purchase and compensation. Its net borrowing is:

$$(P_t + q\tau^p P_t)\epsilon + L(\epsilon) - D_j(\epsilon) = qP_{t+1}\epsilon. \tag{26}$$

The two new bond positions offset. At $t+1$ the buyer sells the extra title for $P_{t+1}\epsilon$ and repays the bond. The old seller's estate receives the same amount through its bond. The timing and quantity of future housing sales, estate expenditures, total tax receipts, and common rebates are unchanged. Existing loans retain their promised repayments. Positive saving provides room to adjust the original financial portfolio; the planner relaxes the private borrowing limit. Uniform strict margins on matched groups permit a common sufficiently small transfer.

With a real cost $C(\epsilon)$, the young owner's exact gain is:

$$\Delta U_i = \log\left(1 - \frac{D_j(\epsilon) + C(\epsilon)}{x_i}\right) + \alpha \log\left(1 + \frac{\epsilon}{s_i}\right). \tag{27}$$

Its derivative at zero is $(MV_i^Y - MV_j^O - K_{ijt})/x_i$. If a fixed cost is present, the finite transfer must keep consumption positive and make this exact gain positive.

A.2 Fertility and a sufficient parameter restriction

For a fixed tenure, positive saving, and unchanged old-age constraints, let $\widetilde{w}$ denote lifetime resources net of any fixed old-age rent. Let $\widehat{h}$ be the current housing limit: $h_R^{\max}$ for a renter and $\min\{h_O^{\max}, b/[(1-\phi)P_t]\}$ for an owner. The conditional problem, after eliminating saving and setting $s = h - \kappa n$, is:

$$\begin{aligned} \max_{x,s,n} \quad & (1+k)\log x + \alpha \log s + \vartheta \log n \\ \text{s.t.} \quad & (1+k)x + us + (\chi + \kappa u)n = \widetilde{w}, \qquad s + \kappa n \le \widehat{h}. \end{aligned} \tag{28}$$

The goods and space used by each child cost $\chi + \kappa u$. For interior old owners, $k = \beta(1+\gamma+\omega_B)$ and $\widetilde{w} = y + b + T_t + qT_{t+1}$. For capped old renters, $k = \beta(1+\omega_B)$ and the old rent must be subtracted:

$$\widetilde{w} = y + b + T_t + qT_{t+1} - q^2 r_{t+1} h_R^{\max}. \tag{29}$$

This reduction requires different formulas if an old owner's retention or estate bound binds.

Differentiating the fertility condition and using $(1+k)\mathrm{d}x = -\chi \mathrm{d}n - p\mathrm{d}h$ gives:

$$\frac{\mathrm{d}n}{\mathrm{d}h} = \frac{\alpha\kappa/s^2 - \chi p/((1+k)x^2)}{\Delta}, \qquad \Delta = \frac{\vartheta}{n^2} + \frac{\chi^2}{(1+k)x^2} + \frac{\alpha\kappa^2}{s^2} > 0. \tag{30}$$

The numerator separates the benefit of additional space from the cost of paying for it.

Differentiating the resource and fertility equations, allowing prices, resources, and the preference weight to change, gives:

$$\Delta\,\mathrm{d}n = \frac{\mathrm{d}\vartheta}{n} + \left(\frac{\alpha\kappa}{s^2} - \frac{\chi u}{(1+k)x^2}\right)\mathrm{d}h + \frac{\chi}{(1+k)x^2}(\mathrm{d}\widetilde{w} - h\,\mathrm{d}u). \tag{31}$$

This separates preference, space, and resource effects at any date. Equation (30) sets the preference weight fixed and collects the net resource change into the payment p.

For a sufficient condition without the borrowing multiplier, write $MV^Y = \alpha x/s$. If $\kappa u/\chi > \alpha/(1+k)$ and the housing cap binds, then $(1+k)\kappa MV^Y > \alpha\chi$. It follows that fertility rises for any $p < MV^Y$. A condition entirely in parameters can imply this inequality.

Proposition 4 (A sufficient parameter restriction)**.** *Suppose property tax and rebates are zero, old-owner choices are interior, saving is positive, the physical owner cap is slack, and the down payment is strictly restrictive. At stationary prices, the following restriction is sufficient for fertility to rise after a local housing increase with net payment $p < MV^Y$:*

$$\frac{(1-q)b}{(1-\phi)(y+b)} \geq \frac{\alpha}{1+\beta(1+\gamma+\omega_B)+\alpha}. \tag{32}$$

It is also sufficient at a transition date if the household expects $P_{t+1} \leq P_t$.

Proof. Let $v = \kappa u/\chi$ and let $a_h = u\widehat{h}/\widetilde{w}$ be the housing expenditure share at the cap. Within the reduced problem, removing the current cap gives the optimal housing expenditure share:

$$g(v) = \frac{\alpha + \vartheta v/(1+v)}{1+k+\alpha+\vartheta}. \tag{33}$$

Strict binding gives $a_h < g(v)$. The function is increasing, and $g(\alpha/(1+k)) = \alpha/(1+k+\alpha)$. At stationary prices, $a_h = (1-q)b/[(1-\phi)(y+b)]$. The restriction therefore implies $v > \alpha/(1+k)$. If $P_{t+1} \leq P_t$, then $u/P_t = 1 - qP_{t+1}/P_t \geq 1-q$, so the same lower bound on a_h holds at that date. The preceding sufficient condition gives the result. □

This parameter restriction excludes very low liquid wealth relative to lifetime resources. More generally, welfare can increase while fertility falls: the exact sign still depends on both space and payment. If a common rental cap increases at both ages, it also raises future rent; under binding caps the appropriate net payment is $p = u_t + qu_{t+1}$.

A lower preference weight weakly lowers an individual household's fertility when its full price and transfer sequence is fixed, even allowing tenure choice. To see this, write its objective as $A(z) + \vartheta \log n(z)$ for a feasible lifetime plan z. Adding the two optimality inequalities at weights ϑ_0 and ϑ_1 gives:

$$(\vartheta_1 - \vartheta_0)(\log n_1 - \log n_0) \geq 0. \tag{34}$$

The feasible set is unchanged. The general-equilibrium sign also requires the price and tenure responses considered below.

A.3 Population units

Let $n^L = a_n n$ count literal children. Consistent rescaling uses $\nu^L = \nu/a_n$, $\chi^L = \chi/a_n$, and $\kappa^L = \kappa/a_n$. Replacement literal fertility is a_n/ν; the change in $\log n$ is a choice-independent constant. If young and old households contain a_Y and a_O adults and children reside with parents in their birth period, resident population is:

$$N_{\mathrm{persons},t} = a_Y Y_t + a_O O_t + a_n \bar{n}_t Y_t. \tag{35}$$

At a positive steady state, $N^*_{\text{persons}}/N^*_{\text{hh}} = (a_Y + a_O + a_n/\nu)/2$. Common fixed counting coefficients therefore preserve the terminal population ranking. Their empirical values are not needed for this illustrative comparison.

B The combined transition

The proof uses the original equilibrium equations on a region where the same household constraints remain strictly binding or slack. Young owners are at their down-payment limits and below their physical caps; young and old renters are at their rental caps. Young households save positively. Old owners have slack retention and estate bounds. All consumption, usable space, fertility, and estate levels are positive. These conditions hold uniformly over the fixed compact support of entrant types.

B.1 Equilibrium equations and inherited claims

Write $u_t = qr_t$, $a = h_R^{\max}$, $\rho_O = 1 + \beta(1 + \gamma + \omega_B)$, and $\rho_R = 1 + \beta(1 + \omega_B)$. Housing choices are $h_i^O = b_i/[(1-\phi)P_t]$ and $h_i^R = a$. The conditional resource equations and fertility conditions are:

$$\begin{aligned}
\rho_O x_i^O + \chi n_i^O &= w_i - u_t h_i^O, \\
\rho_R x_i^R + \chi n_i^R &= w_i - u_t a - q u_{t+1} a, \\
\frac{\vartheta}{n_i^m} &= \frac{\chi}{x_i^m} + \frac{\alpha\kappa}{h_i^m - \kappa n_i^m}, \qquad w_i = y_i + b_i + T_t + qT_{t+1}.
\end{aligned} \tag{36}$$

The future rent affects renter choices and therefore tenure probabilities. For the cohorts choosing under these prices, old-owner housing is $\beta\gamma x_i^O/(qu_{t+1})$ and old-renter housing is a.

The aggregate state is $Z_t = (P_t, u_t, Y_t, O_t, M_t, R_t)$, where M_t/u_t and R_t are total old-owner and old-renter housing. Given this state, rental entry determines P_{t+1} and fiscal balance determines T_t. The two remaining unknowns $v = (u_{t+1}, Y_{t+1})$ solve:

$$\begin{aligned}
F_1(Z_t, v) &= Y_t \int [\pi_i^O h_i^O + (1-\pi_i^O)a] \, \mathrm{d}F + M_t/u_t + R_t - \bar{H} = 0, \\
F_2(Z_t, v) &= Y_{t+1} - \nu Y_t \int [\pi_i^O n_i^O + (1-\pi_i^O) n_i^R] \, \mathrm{d}F = 0.
\end{aligned} \tag{37}$$

These are housing clearing and the cohort law. The next state is:

$$G(Z_t, v) = \left(P_{t+1}, u_{t+1}, Y_{t+1}, Y_t, \frac{\beta\gamma}{q} Y_t \int \pi_i^O x_i^O \, \mathrm{d}F, a Y_t \int (1-\pi_i^O) \, \mathrm{d}F\right). \tag{38}$$

If F_v is nonsingular, these equations give a smooth local map $Z_{t+1} = g(Z_t; \vartheta, \phi)$. It is derived from the original equilibrium, including old-state evolution.

At an unexpected policy date, let B, A, H denote the inherited old-owner mass, net financial claims, and purchased titles, respectively. Let $\mathcal{I} = (Y_{t_p}, O_{t_p}, B, A, H)$ and $L = \gamma/(1 + \gamma + \omega_B)$. Initial housing demand satisfies:

$$Y = Y_{t_p}, \quad O = O_{t_p}, \quad R = a(O_{t_p} - B), \quad M = L[A + PH + T(P, Y, O)B]. \tag{39}$$

The title price and rebate can change, so M is not predetermined wealth. Financial claims include the original mortgage repayment; the new ϕ does not rewrite that debt. The full conditional inherited states are also retained to check each old household's feasibility.

Previously chosen saving and rent use the forecasts held before the shock. In particular, rental entry at $t_p - 1$ uses the baseline expected price at t_p, and the old cohort's earlier saving condition need not hold at the surprisingly realized price. Equations (36)– (38) apply to the new continuation from t_p onward, with (39) determining the initial old.

B.2 Local continuation and the two shocks

Let Z^* be a positive stationary equilibrium satisfying the stated strict household conditions. Assume g and the four initial restrictions b are smooth, F_v is nonsingular, and $J = g_Z(Z^*)$ has four eigenvalues strictly inside and two strictly outside the unit circle. Let $B_0 = b_Z$ at the initial stationary data. Its restriction to the stable generalized eigenspace must be invertible. Finally, at this equilibrium the impact fertility and stationary population derivatives with respect to both ϑ and ϕ are strictly positive, holding inherited claims fixed in the impact comparisons.

Here is the local argument. Write $\lambda = (\vartheta, \phi)$ for the two parameters. Since one is not an eigenvalue of J, the stationary state $Z^*(\lambda)$ is smooth. Center a continuation on it by $v_k = Z_{t_p+k} - Z^*(\lambda)$. The exact sequence equations are:

$$b(Z^*(\lambda) + v_0; \mathcal{I}) = 0, \qquad v_{k+1} - g(Z^*(\lambda) + v_k; \lambda) + Z^*(\lambda) = 0. \tag{40}$$

Centering removes the permanent parameter shift from the tail. Choose $\eta < 1$ above the stable spectral radius and use the sequence norm $\sup_k \eta^{-k}\|v_k\|$. The derivative in v is $(B_0 v_0, \{v_{k+1} - J v_k\})$. For a forcing f_k, its stable and unstable components are solved by:

$$v_k^s = J_s^k c + \sum_{j=0}^{k-1} J_s^{k-1-j} f_j^s, \qquad v_k^u = -\sum_{j=k}^{\infty} J_u^{k-1-j} f_j^u. \tag{41}$$

The boundary determines c uniquely; both sums are bounded in this norm. Only the unstable block is inverted, so a zero stable root is allowed. Uniform household margins make the residual continuously differentiable. The sequence implicit-function theorem therefore gives a smooth, exponentially converging continuation for nearby parameters and inherited data. The same inverse on bounded sequences supplies uniqueness among continuations staying in a sufficiently small common neighborhood.

For the initial preference decline, inherited data are the original stationary data. Smoothness gives a constant C such that:

$$\|Z_t^0(\epsilon) - Z^*(\vartheta_0 - \epsilon, \phi_0)\| \le C\epsilon\eta^t, \qquad \|Z^*(\vartheta_0 - \epsilon, \phi_0) - Z^*(\vartheta_0, \phi_0)\| \le C\epsilon. \tag{42}$$

The entire baseline stays uniformly close to the original equilibrium. The strict preference derivatives give lower initial fertility and a smaller endpoint for sufficiently small positive ϵ.

Original household budgets reconstruct assets, titles, and tenure probabilities smoothly on the fixed compact type support. Their deviations are therefore uniformly of order ϵ, across every date and type. This controls individual feasibility as well as the aggregate inherited condition; closeness of aggregate moments alone would not suffice. At $\delta = 0$, local uniqueness identifies the continuation at t_p with the baseline tail. Continuity preserves the strict policy derivatives for all these inherited states in one neighborhood. Integrating them from ϕ_0 to $\phi_0 + \delta$ gives (23). The uniform bound permits common positive upper bounds on ϵ and δ, independent of the chosen date t_p. This proves Proposition 3 once an economy meeting the strict conditions is supplied.

B.3 An economy satisfying the conditions

Use the following primitive values, with homogeneous income and liquid wealth and a finite ownership-taste scale:

$$\begin{aligned} &q = \tfrac{1}{2}, \quad \phi = \tfrac{4}{5}, \quad b = \tfrac{1}{5}, \quad \alpha = \beta = \omega_B = \tfrac{2}{5}, \quad \gamma = \tfrac{3}{10}, \\ &\chi = \tfrac{3}{20}, \quad \kappa = \tfrac{1}{2}, \quad \vartheta = \tfrac{141}{400}, \quad \nu = 2, \quad h_R^{\max} = \tfrac{1}{4}, \quad h_O^{\max} = 2, \quad \sigma_\xi = 4, \\ &\tau^p = \tfrac{467}{9250}, \quad y = \tfrac{3521349281}{1677802000}, \quad \bar{H} = \tfrac{68104}{68019}. \end{aligned} \tag{43}$$

Set $\bar{\xi} = \sigma_\xi \log(\pi/(1-\pi)) - W^O + W^R$ at $\pi = 11/21$ and the allocation below, then hold both taste parameters fixed during the shocks. This specifies one preference distribution.

There is an exact stationary equilibrium with $P = Y = O = 1$, $u = 9717/18500$, $T = 3975571/314587875$, and:

$$(x_O, h_O, n_O) = (1, 1, \tfrac{3}{4}), \qquad (x_R, h_R, n_R) = (\tfrac{99}{74}, \tfrac{1}{4}, \tfrac{9}{40}). \tag{44}$$

Mean fertility is 1/2 and 11/21 of households own. The young owner's housing value is 16/25, exceeding the old owner's value 9717/18500. All stated household inequalities have strict margins. Thus the allocation and transition results apply in the same economy.

Differentiating (36) gives the two additional terms needed for the preference shock: $\mathrm{d}\vartheta/n_i^m$ in the fertility condition and $\log n_i^m \, \mathrm{d}\vartheta$ in the conditional value. Together with (12), the latter accounts for endogenous tenure changes. Let $Q_a = g_a$ for $a \in \{\vartheta, \phi\}$; then:

$$J = G_Z - G_v F_v^{-1} F_Z, \qquad Q_a = G_a - G_v F_v^{-1} F_a, \qquad Z_a^* = (I - J)^{-1} Q_a. \tag{45}$$

These expressions follow directly from the two implicit equations.

Exact root bounds give one zero root, one real stable root in $(.00319, .00320)$, a nonreal pair with modulus in $(.41, .43)$, and unstable roots in $(-80.93, -80.92)$ and $(4.148, 4.149)$. The four initial restrictions are independent on the stable space. Solving the linear recurrence and those restrictions gives:

Derivative at the reference equilibrium	ϑ increases	ϕ increases
Initial mean fertility	0.84713 to 0.84714	0.07966 to 0.07968
Stationary adult households	4.79728 to 4.79729	2.33786 to 2.33788

The bounds are outward enclosures; all four derivatives are positive. The preference derivative uses $1 < \log(10/3) < 2$, with a tighter rational series bound for the displayed intervals. Strict inequalities persist in an open neighborhood. Uniformly small income and wealth variation on a fixed compact probability space preserves the argument as well. This does not impose zero renter mass, zero child goods costs, or zero property tax.

The local argument above does not give a numerical radius. At the reference economy, a separate proof with verified interval bounds establishes the transition and sign comparisons for $0 < \epsilon \le 1/20000$ and $0 < \delta \le 1/20000$, uniformly over $t_p \ge 1$. The bounds apply to both infinite paths and preserve the stated household constraints. This is a small range: the financed share rises from 80% to at most 80.005%. The proof and reproducible checks are recorded in the supporting verification note. This bound concerns the reference economy; it is not a common numerical radius for every economy in the open set. Larger shocks can change which constraints bind or eliminate the positive endpoint. For example, the fertility condition implies:

$$n < \frac{\vartheta h}{\kappa(\alpha + \vartheta)} \le \frac{\vartheta h_O^{\max}}{\kappa(\alpha + \vartheta)}. \tag{46}$$

If the last bound is at most $1/\nu$, a positive closed steady state is impossible. The local construction does not cover such a decline.

B.4 The two-panel comparison

Figure 2 uses the analytical first-order response at the economy above. Take a preference decline of size ϵ, introduce the credit change at $t_p = 1$, and set $\delta = \epsilon/2$. These choices specify a direction as

ϵ tends to zero; they do not select a finite reform size or a historical date. If z_k^a is the first-order response to a permanent unit increase in parameter a from the reference equilibrium, the two displayed directions are:

$$z_t^0 = -z_t^{\vartheta}, \qquad z_t^1 = \begin{cases} z_t^0, & t < 1, \\ z_t^0 + \frac{1}{2} z_{t-1}^{\phi}, & t \geq 1. \end{cases} \tag{47}$$

Their interaction is of second order. The policy response at date 1 satisfies the homogeneous version of (39), so it preserves that date's inherited cohorts and claims.

The dashed lines are tangents to housing and fertility schedules evaluated at constant prices and equal age-cohort masses. Write $N = N_{\mathrm{hh}}$. At a candidate (P, N), set $T = q\tau^p P\bar{H}/N$, solve the household problems with repeated prices, and define:

$$\mathcal{H}(P, N; \vartheta, \phi) = \frac{N}{2}(\bar{h}^Y + \bar{h}^O) - \bar{H}. \tag{48}$$

The left panel solves $\mathcal{H} = 0$ locally for price as a function of population; the right panel evaluates mean fertility on that same schedule. On the underlying exact schedules, an intersection with replacement fertility is a stationary equilibrium. The drawn tangent intersections give its first-order location. Other points on the schedules need not satisfy the demographic law.

The solid arrows connect selected transition equilibria. During adjustment, cohort proportions, inherited assets, and expected future prices differ from the stationary schedules, so a transition point need not lie on one of their dashed lines. The arrows compare initial, intervention-date, and limiting states; they do not impose monotone motion between them. Displayed coordinates are first-order changes per unit ϵ from the old equilibrium; symbolic labels refer to economic levels. Both axes use affine scales.